

Karam Thethy

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TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, HTML/CSS

Backend: Spring Boot, Django, FastAPI, Kafka, PostgreSQL, Redis, WebSockets, gRPC, REST APIs

Cloud & DevOps: Docker, Kubernetes, Azure DevOps, ArgoCD, GitHub Actions, OpenTelemetry

Frontend: React, Next.js, Vue 3, Tailwind CSS

Tools: Git, Jira, Datadog

PROFESSIONAL EXPERIENCE

Software Engineer (Graduate Program)

Feb. 2025 – Present

Coles

Melbourne, VIC

Backend Engineer

Feb. 2026 – Present

- Maintaining Java Spring Boot microservices on Kubernetes powering Coles Online cart, payment, and order fulfilment flows, processing Kafka event streams across distributed services including payment vaults, order management, and warehouse systems.
- Built **RewriteBot**, a self-hosted Azure DevOps pipeline system automating OpenRewrite code transformations fleet-wide — scanner runs weekly producing patch files and reports across all repos; bot runs on-demand with dry-run gating, `mvn verify` checks, and automated PR creation via Azure CLI.
- Identified and reported 5 upstream bugs in the OpenRewrite open-source project during Spring Boot 4 migration, including a behaviour-changing semantic rewrite of `@ConditionalOnBean` annotation handling; contributed isolated reproduction modules for each issue.

Frontend Engineer

Jul. 2025 – Feb. 2026

- Designed and built a Chrome DevTools extension (Manifest V3, vanilla JS) automating barcode scan simulation for a React/TypeScript PWA used across ~800 stores, removing a critical QA bottleneck for the frontend team.

Full Stack Developer (Contractor)

Nov. 2023 – Mar. 2024

Department of Fire and Emergency Services (DFES) WA

- Architected the Django backend for a government proof-of-concept emergency platform — designed REST APIs, PostgreSQL schemas for chat threads and geospatial incident data, real-time WebSocket streaming via Redis Channels, and OpenAI integration for context-aware evacuation plan generation.
- Containerised the full stack with Docker Compose behind Nginx with Let's Encrypt SSL, managed CI/CD through GitHub Actions, and maintained separate dev/prod environment configurations for a 4-person team delivering over 4 months of sprints.

Research Officer

Jul. 2023 – Apr. 2025

The University of Western Australia

- Built a RAG pipeline integrating LLMs with vector search across a 350,000+ concept SNOMED-CT ontology to automate structured coding of unstructured ophthalmology clinical letters.
- Developed a stakeholder-facing Django application for clinicians to evaluate and validate automated coding outputs against ground-truth annotations.

PROJECTS

Odyssey ROV HMI | *React 19, FastAPI, WebSockets, gRPC, Protocol Buffers, Docker*

- Designed and built a full-stack real-time Human-Machine Interface for a simulated deep-sea ROV, featuring a custom Python simulation engine with physics-based telemetry, stateful scenario logic, and a discrete mission state machine.
- Implemented a dual-protocol architecture — WebSocket for 2 Hz bidirectional telemetry and gRPC for structured historical queries — with strict Pydantic data contracts shared across the full stack.

Samsung Solve for Tomorrow (Panopticon) | *SvelteKit, Mapbox, Chart.js, Tailwind CSS*

2024

- Won national first prize (\$10,000) for an agricultural IoT dashboard enabling data-driven pesticide deployment through automated insect surveillance.
- Built a reactive SvelteKit frontend with Mapbox heatmap overlays, per-sensor time-series charting, and geospatial crop field visualisation across satellite imagery.

EDUCATION

The University of Western Australia

Bachelor of Philosophy (Honours) in Computer Science, First Class Honours

WAM: 83.4 / 100.0 GPA: 6.7 / 7.0

Feb. 2020 – Dec. 2023

Perth, WA